CFG to PDA

Step 1 : Remove Null Productions

Step 2 : Remove unit Productions

Step 3 : Obtain GNF / convert in GNF

Every production must begin with a terminal:

$A \rightarrow a\alpha$

where: a is a terminal

α is a sequence of non terminals

Step 4 : Transition of PDA

1) For all CFG Productions

$A \rightarrow B$ here, A = left symbol

$\delta(q, \epsilon, A) = (q, B)$ β = right production

2) For all terminals

$a \in T$ here, a = terminal

$\delta(q, a, a) = (q, \epsilon)$

ex: CFG / Grammar $= (\{S, A\}, \{0, 1\}, P, S)$

P is defined by :

$S \rightarrow 0S \mid A$

$A \rightarrow 1A0 \mid S \mid \epsilon$

→ **Step 1 :** Remove Null Productions

in this grammar, 2 null productions

$A \rightarrow \epsilon$

$\underbrace{S \rightarrow A}_{\text{why ?}}$

because, if we put $A \rightarrow \epsilon$ in $S \rightarrow A$,

it will be $S \rightarrow \epsilon$.

if we put ϵ in place of S & A,

1) $S \rightarrow 0S$
↓
$S \rightarrow 0\epsilon$

2) $S \rightarrow 0$

3) $S \rightarrow A$
↓
$S \rightarrow \epsilon$

1) $A \rightarrow 1A0$
↓
$A \rightarrow 1\epsilon 0$

2) $A \rightarrow 10$

3) $A \rightarrow S$
↓
$A \rightarrow \epsilon$

→

$S \rightarrow 0S \mid 0 \mid A$

$A \rightarrow 1A0 \mid 10 \mid S$

Step 2 : Remove unit Productions

$S \to A$ } $\to$ unit productions
$A \to S$

→ replace A with its productions

$\therefore \; S \to 0S \mid 0 \mid 1A0 \mid 10 \mid S$

$A \to 1A0 \mid 10 \mid S$

→ here, useless symbol is $S \to S$, so remove it.

→ replace S with its productions, in $A \to S$
bcz $A \to S$ is unit production

$\therefore \; S \to 0S \mid 0 \mid 1A0 \mid 10$

$A \to 1A0 \mid 10 \mid 0S \mid 0 \mid 1A0 \mid 10$

→ remove duplicates :

$S \to 0S \mid 0 \mid 1A0 \mid 10$

$A \to 1A0 \mid 10 \mid 0S \mid 0$

So, grammar after removing unit productions, duplicates :

$S \to 0S \mid 0 \mid 1A0 \mid 10$

$A \to 0S \mid 0 \mid 1A0 \mid 10$

Step 3 : Obtain GNF / convert in GNF

GNF rule : every production must be of form :

$A \to a\alpha$

where, a = terminal
α = sequence of non-terminals.

→ in the above given grammar,
two productions are not in GNF

$S \to 1A0$

$S \to 10$

So, introduce new Non-terminal / variable

$B \to 0$

now replace : $1A0 \to 1AB$

$10 \to 1B$

final GNF Grammar :

$S \to 0S \mid 0 \mid 1AB \mid 1B$

$A \to 0S \mid 0 \mid 1AB \mid 1B$

$B \to 0$

Step 4 : Transition of PDA (Answer)

1) For all CFG Productions

$A \to B$

$\delta(q, \epsilon, A) = (q, B)$

here, A = left symbol
B = right production

① $S \to 0S$

$\delta(q, \epsilon, S) = (q, 0S)$

② $S \to 1AB$

$\delta(q, \epsilon, S) = (q, 1AB)$

③ $S \to 1B$

$\delta(q, \epsilon, S) = (q, 1B)$

④ $S \to 0$

$\delta(q, \epsilon, S) = (q, 0)$

⑤ $B \to 0$

$\delta(q, \epsilon, B) = (q, 0)$

⑥ $A \to 0S$

$\delta(q, \epsilon, A) = (q, 0S)$

⑦ $A \to 1AB$

$\delta(q, \epsilon, A) = (q, 1AB)$

⑧ $A \to 1B$

$\delta(q, \epsilon, A) = (q, 1B)$

⑨ $A \to 0$

$\delta(q, \epsilon, A) = (q, 0)$

2) For all terminals

$a \in T$

$\delta(q, a, a) = (q, \epsilon)$

here, a = terminal

in our CFG, there are two terminals : 0, 1

① $\delta(q, 0, 0) = (q, \epsilon)$

② $\delta(q, 1, 1) = (q, \epsilon)$